Operating Systems Notes

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1 OS Fundamentals

OS provides interface between user and hardware. Program divided into pages, OS into frames. Frame size $\geq$ Page size. CPU generates logical address (Page no — Page offset). Virtual memory used when main memory $<$ program size.

Page Table: Stores frame number and validity. Access time = 2m.

2 Types of Kernel

1. Monolithic: Fast, less stable 2. Microkernel: Reliable, slower 3. Hybrid: Combination 4. Exokernel: Direct hardware access 5. Nano: Minimal services

3 Types of OS

Batch: Sequential execution, no priority **Multiprogramming:** Multiple jobs, switches on I/O **Multitasking:** Time sharing **Multiprocessor:** Multiple CPUs **Real-time:** Deadline-driven **Distributed:** Network resources

4 Process Management

Process = Program in execution. PCB stores: ID, state, counter, priority, registers, files, devices, protection.

States: New $\rightarrow$ Ready $\rightarrow$ Running $\rightarrow$ Waiting $\rightarrow$ Terminate

5 Key Concepts

System call: OS service (kernel mode) Library function: Compiler service (user mode) Booting: Loading kernel Spooler: FIFO buffer for devices